ArchAnalyse

Architectural Drawing Intelligence System

User Manual

A Collaborative Team Project

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1. Introduction

ArchAnalyse is a web-based Architectural Drawing Intelligence System designed to assist structural engineers in reviewing and analysing floor plan drawings. The system provides the following capability:

• **Wall Segmentation** — applies a deep-learning model (SegFormer-B2) to identify and highlight wall regions in floor plan images.

This manual covers system requirements, installation, account management, and step-by-step instructions for every feature available in the application.

2. System Requirements

2.1 Minimum Requirements

Component	Requirement
Operating System	Windows 10/11, macOS 12+, or Ubuntu 20.04+
RAM	8 GB (16 GB recommended for wall segmentation)
Disk Space	At least 5 GB free (model weights ~100 MB included)
Docker	Docker Desktop 4.x or Docker Engine 24+
Browser	Google Chrome 100+, Firefox 100+, or Edge 100+

2.2 GPU Support (Optional)

Wall segmentation runs significantly faster with a compatible NVIDIA GPU. The startup script detects GPU availability automatically via nvidia-smi — if a CUDA-capable GPU is found the system builds with GPU support, otherwise it falls back to CPU-only mode.

> ■ **Note**
> GPU support requires CUDA 11.8.

Three components must be installed on the host machine before starting Docker. Follow the steps for your operating system.

Windows

1. Install NVIDIA GPU Drivers — download and run the latest driver for your GPU from https://www.nvidia.com/Download/index.aspx, then verify with: nvidia-smi

2. Install CUDA Toolkit 11.8 — download the Windows exe installer from https://developer.nvidia.com/cuda-11-8-0-download-archive (select Windows → x86_64 → exe local) and run it. Verify with: nvcc --version

3. Enable GPU in Docker Desktop — Docker Desktop on Windows uses WSL 2 to expose GPUs to containers. No separate NVIDIA Container Toolkit is needed. Go to Docker Desktop → Settings → Resources → WSL Integration and ensure your WSL 2 distro is enabled. Verify Docker can see the GPU:

```
docker run --rm --gpus all nvidia/cuda:11.8.0-base-ubuntu22.04 nvidia-smi
# Expected: same nvidia-smi table as on the host
```

> ■ **Windows Requirement**
>
> GPU passthrough requires WSL 2 and Docker Desktop 4.x or later.
> The legacy Hyper-V backend does not support GPU passthrough.

Linux (Ubuntu 20.04 / 22.04)

4. Install NVIDIA GPU Drivers:

```
sudo apt-get update && sudo apt-get install -y ubuntu-drivers-common
sudo ubuntu-drivers autoinstall && sudo reboot
nvidia-smi   # verify after reboot
```

5. Install CUDA Toolkit 11.8:

```
wget https://developer.download.nvidia.com/compute/cuda/11.8.0/local_installers/\
  cuda_11.8.0_520.61.05_linux.run
sudo sh cuda_11.8.0_520.61.05_linux.run --silent --toolkit
echo 'export PATH=/usr/local/cuda-11.8/bin:$PATH' >> ~/.bashrc
echo 'export LD_LIBRARY_PATH=/usr/local/cuda-11.8/lib64:$LD_LIBRARY_PATH' >> ~/.bashrc
source ~/.bashrc && nvcc --version
```

6. Install NVIDIA Container Toolkit:

```
curl -fsSL https://nvidia.github.io/libnvidia-container/gpgkey \
  | sudo gpg --dearmor -o /usr/share/keyrings/nvidia-container-toolkit-keyring.gpg
curl -s -L https://nvidia.github.io/libnvidia-container/stable/deb/nvidia-container-toolkit.
list \
  | sed 's#deb https://#deb [signed-by=...] https://#g' \
  | sudo tee /etc/apt/sources.list.d/nvidia-container-toolkit.list
sudo apt-get update && sudo apt-get install -y nvidia-container-toolkit
sudo nvidia-ctk runtime configure --runtime=docker
sudo systemctl restart docker
docker run --rm --gpus all nvidia/cuda:11.8.0-base-ubuntu22.04 nvidia-smi
```

macOS

> ■ **macOS — No CUDA Support**
>
> Apple Silicon and Intel Macs do not support NVIDIA CUDA.
> ArchAnalyse runs in CPU-only mode on macOS automatically. No additional installation is required.

3. Installation & Starting the Application

3.1 Prerequisites

Before starting, ensure the following are installed on your machine:

• Docker Desktop (Windows / macOS) or Docker Engine + Docker Compose (Linux)

• NVIDIA drivers, CUDA Toolkit 11.8, and NVIDIA Container Toolkit — only required for GPU acceleration (see Section 2.2)

3.2 Starting on Windows

Double-click **start.bat** in the project root folder. An interactive menu appears:

```
============================================
  ArchAnalyse - Startup
============================================

How would you like to run ArchAnalyse?

  [1] Auto-detect  (use GPU if available, otherwise CPU)
  [2] GPU only     (force GPU mode - requires NVIDIA CUDA)
  [3] CPU only     (force CPU mode)

Enter choice [1/2/3] (default: 1):
```

Enter 1, 2, or 3 and press Enter. Pressing Enter with no input defaults to auto-detect.

Option	Mode	Behaviour
1	Auto-detect	Checks nvidia-smi; uses GPU if found, otherwise CPU.
2	GPU only	Forces GPU mode. Shows a warning if no GPU is detected.
3	CPU only	Forces CPU mode regardless of hardware.

3.3 Starting on macOS / Linux

```
chmod +x start.sh
./start.sh
```

The same interactive menu is shown. Enter your choice and press Enter.

3.4 Accessing the Application

Once the containers are running, open your browser and navigate to:

■ Application URL	**http://localhost**

The backend API is accessible at http://localhost:8000

3.5 Stopping the Application

```
docker compose down
```

4. Account Management

4.1 User Registration

New users must register for an account before using the system. Accounts require administrator approval before login is permitted.

1. Navigate to http://localhost in your browser.

2. Click **Register** on the login page.

3. Enter a Username (minimum 3 characters), a valid Email address, and a Password (minimum 6 characters).

4. Click **Submit**. You will see a confirmation message: "Registration successful. Please wait for admin approval before logging in."

5. Contact your system administrator to approve your account.

> ■ **Important**
> You cannot log in until an administrator approves your account.
> Contact your admin if you have not received approval within a reasonable time.

4.2 Logging In

6. Go to http://localhost.

7. Enter your Username and Password.

8. Click **Log In**.

9. Upon successful login, you are directed to the main dashboard.

4.3 Default Administrator Account

A default administrator account is created automatically when the system first starts:

Field	Default Value
Username	admin
Password	admin123
Email	admin@archanalyse.com

> ■ **Security Notice**
> Change the default administrator password immediately after first login.
> The default credentials can be overridden via environment variables ADMIN_USERNAME, ADMIN_PASSWORD, and ADMIN_EMAIL before starting the containers.

5. Administrator Panel

The administrator panel is accessible to accounts with admin privileges. It allows administrators to review and manage user registrations.

5.1 Approving Pending Users

10. Log in with an administrator account.

11. Navigate to the **Admin Panel** from the top navigation bar.

12. The **Pending Approvals** section lists all users who have registered but not yet been approved.

13. Click **Approve** next to a user to grant them access. The user can now log in.

5.2 Managing Existing Users

The **All Users** section displays every registered account. Administrators may delete user accounts from this view. Deleting an account is permanent and cannot be undone.

6. Wall Segmentation

The Wall Segmentation feature uses a SegFormer-B2 deep-learning model trained on architectural floor plans to identify wall regions in one or more images. Results are shown as colour overlays and isolated wall extractions.

6.1 Running Wall Segmentation

14. Click the **Wall Segmentation** tab in the top navigation bar.
15. Click **Choose Images** and select one or more floor plan image files (.png, .jpg).
16. Click **Run Segmentation**. A progress bar shows the processing status for each image.
17. Once all images are processed, thumbnails appear in the results panel.

6.2 Viewing Results

Each processed image has two views:

View	Description
Overlay	The original image with detected wall regions highlighted in red.
Extracted Walls	Only the wall pixels are shown; all other areas are masked out.

Click the **Download** button under either view to save the image to your computer.

6.3 Processing Details

• Images are resized to a maximum of 1024 px on the long edge before inference.

• Each image is pre-processed with Otsu binarisation to match the model's training format.

• Batch processing streams progress updates in real time.

7. Segmentation History

ArchAnalyse saves the results of past wall segmentation sessions so you can review previous analyses without re-running the model.

7.1 Viewing History

18. Click the **History** tab in the top navigation bar.

19. A list of previous segmentation sessions is displayed, showing the file name, date, and a thumbnail preview.

20. Click any entry to view the original image, the overlay, and the extracted walls.

7.2 Exporting History Results

From a history entry detail view, click **Export PDF** to generate a PDF report for that segmentation result. The report can then be downloaded.

7.3 Deleting History Entries

To delete a history record, click the **Delete** button next to the entry in the history list. Deletion is permanent.

8. Troubleshooting

Problem	Solution
Cannot access http://localhost	Ensure Docker containers are running: docker compose ps. Check that port 80 is not in use by another application.
Login says 'pending approval'	Contact your administrator to approve your account in the Admin Panel.
Wall segmentation fails	Ensure the image is a valid .png or .jpg file. Very small images (< 64 px) may not be supported.
GPU not being used	Verify that NVIDIA drivers and the NVIDIA Container Toolkit are installed. Run nvidia-smi in a terminal to confirm GPU visibility.
Docker build fails	Ensure you have a stable internet connection for downloading dependencies. Try running docker compose build --no-cache to force a clean rebuild.

9. API Reference (Advanced Users)

The backend exposes a REST API at http://localhost:8000. Developers and advanced users can interact with it directly.

Method	Endpoint	Description
GET	`/health`	Health check — returns system status
POST	`/api/auth/register`	Register a new user account
POST	`/api/auth/login`	Obtain a JWT access token

Method	Endpoint	Description
GET	`/api/auth/me`	Get the current authenticated user's details
POST	`/api/segment`	Segment walls in a single image
POST	`/api/segment-batch`	Batch segmentation with streaming progress
GET	`/api/history`	List all segmentation history records
DELETE	`/api/history/{record_id}`	Delete a specific history record
GET	`/api/history/{record_id}/pdf`	Export a history record as PDF
POST	`/api/export-segment-pdf`	Export a live segmentation result as PDF
GET	`/api/admin/pending`	List users awaiting approval (admin only)
POST	`/api/admin/approve/{user_id}`	Approve a pending user (admin only)
DELETE	`/api/admin/users/{user_id}`	Delete a user account (admin only)

Interactive API documentation (Swagger UI) is available at http://localhost:8000/docs

10. File Storage Locations

Files created by the system are stored inside the Docker volumes, which map to the following host directories:

Location	Contents
./backend/history_images/	Saved segmentation result images (overlay & extracted)

This directory persists between container restarts. To clear all stored data, delete the contents of this folder and restart the containers.

11. Glossary

Term	Definition
SegFormer-B2	A transformer-based semantic segmentation model used to identify wall pixels in floor plan images.
Otsu Binarisation	An image pre-processing technique that converts a grayscale image to black-and-white by automatically finding the optimal threshold.
mIoU	Mean Intersection over Union — a standard metric for evaluating segmentation accuracy.

Term	Definition
JWT	JSON Web Token — a compact token format used for authenticating API requests.
Docker Compose	A tool for defining and running multi-container Docker applications.
Overlay	A visualisation where detected wall regions are highlighted in red over the original floor plan image.
Extracted Walls	A visualisation showing only the detected wall pixels; all other regions are masked to black.

End of User Manual — ArchAnalyse v1.0